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Thirteen-Velocity Three-Dimensional Multiple-Relaxation-Time Lattice Boltzmann Model for Incompressible Navier-Stokes Equations

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Abstract. This paper presents a three-dimensional thirteen-velocity (D3Q13) multiple-relaxation-time (MRT) lattice Boltzmann (LB) model for incompressible flows. The model can eliminate the second kind of compressibility error existing in previous D3Q13 MRT LB models. We simulate the three-dimensional steady Poiseuille flow, unsteady pulsatile flow, and lid-driven cavity flow to verify the validity of the current model. The excellent agreement between the numerical result and the analytical solution or the previous numerical result demonstrates the model's effectiveness.

AMS subject classifications: 76M25, 76D05

Key words: Lattice Boltzmann method, multiple relaxation time, incompressible Navier-Stokes equations, compressibility error.

1 Introduction

In the past decades, researchers have accepted the LB method as an effective numerical method for solving incompressible Navier-Stokes (N-S) equations [1–3]. Compressibility error is considered an essential issue in solving incompressible N-S equations by the LB model [4–8]. However, few papers have been published to elucidate the compressibility

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error occurring in the LB method [9–12]. Generally, the compressibility error can be attributed to the errors of two kinds. The first is the error caused by the non-zero Mach number. For example, the accuracy of the macroscopic momentum equations of most LB models is of order $\mathcal{O}(\delta t^2+\delta t M^2)$, where δt denotes the time step and M denotes the Mach number. If the Mach number is not zero, the error such as $\mathcal{O}(\delta t M^2)$ always exists. Essentially, this error is caused by the compressibility of the flow, so it can be named the compressibility error of the first kind. The deviation of the macroscopic equations of the LB models from the incompressible N-S equations causes the second kind of compressibility error. Under the low Mach number assumption, we expect an LB model to recover to the incompressible N-S equations because the low Mach number assumption is equivalent to the incompressible limit. Unfortunately, most LB models can only recover to the compressible N-S equations under the low Mach number assumption. When we use these LB models to simulate incompressible flows, the LB solution may deviate from the solution of the incompressible N-S equations. This deviation is called the compressibility error of the second kind.

In the past, various efforts have focused on the second kind of compressibility error [4–8]. For example, Guo et al. [4] proposed a two-dimensional nine-velocity (D2Q9) lattice Bhatnagar-Gross-Krook (LBGK) model, which can eliminate the compressibility error of the second kind in the existing two-dimensional (2D) LBGK models [12, 13]. He et al. [5] presented a unified DdQq incompressible LBGK model for 2D and three-dimensional (3D) incompressible flows, which can eliminate the compressibility error of the second kind in the existing 3D LBGK models. Based on the equilibrium distribution functions of Guo et al. D2Q9 LBGK model, Du et al. [6] and Du [7] developed new versions of D2Q9 and D3Q19 MRT LB models for 2D and 3D incompressible flows, respectively. Zhang et al. [8] proposed an LBGK D2Q9 model for incompressible axisymmetric flows, which can eliminate the compressibility error of the second kind in the existing LB models for axisymmetric flows.

In the LB method, the commonly used 2D and 3D models are D2Q9, D3Q13, D3Q15, D3Q19 and D3Q27 models [13–16]. Generally, the MRT LB model is much more stable than the LBGK model. However, the computational speed of the MRT LB model in terms of the number of nodes updated per second is about 15% slower than that of the LBGK model [15]. To realize a computationally more efficient two-dimensional MRT LB model for incompressible flows, Du et al. proposed a D2Q8 MRT LB model [17]. Enlightened by Du et al.'s work, Zhang et al. proposed D3Q14 and D3Q18 MRT LB models for three-dimensional incompressible flows [18].

To facilitate the discussion, we refer to all the above-discussed LB models as the scalar LB models [19]. In the 3D space, a scalar LB model usually needs at least 13 discrete velocity directions to solve the incompressible N-S equations. Following the developments of the D2Q8, D3Q14, and D3Q18 MRT LB models, an intuitive question is if a D3Q12 MRT LB model can be constructed similarly. This realization could be instrumental because a D3Q12 MRT LB model is, in principle, the computationally most efficient 3D MRT LB model. However, constructing such a model is challenging because the method adopted for constructing the D2Q8, D3Q14, and D3Q18 MRT LB models cannot be directly used.

Recently, we proposed an inversion method to overcome the above-discussed difficulty, and by using this method, we proposed a D3Q12 MRT LB model for 3D incompressible flows [20]. In such a development, we found that by using the inversion method, we can also obtain a new D3Q13 MRT LB model, which differs from the existing D3Q13 MRT LB models [14]. More importantly, this subsequent D3Q13 MRT LB can recover to the exact 3D incompressible N-S equations only under the low Mach number assumption. This D3Q13 MRT LB model can eliminate the compressibility error of the second kind, which has yet to be achieved by the developed D3Q13 MRT LB models.

The rest of the paper is organized as follows. Section 2 introduces the existing D3Q13 MRT LB models. Section 3 presents a new D3Q13 MRT LB model and describes the methods to construct this model. We perform three benchmark tests to validate the current D3Q13 MRT LB model in Section 4. Finally, Section 5 gives the conclusion.

2 The existing D3Q13 MRT LB models

The two existing D3Q13 MRT LB models are shown in [14]. We call them the D3Q13 MRT LB model and the "incompressible" D3Q13 MRT LB model in this paper. The existing D3Q13 MRT LB models adopt discrete velocities as follows

$$\{\boldsymbol{c}_0,\boldsymbol{c}_1,\ldots,\boldsymbol{c}_{12}\}=\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 & -1 & -1 & -1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 & 1 & -1 & 1 & 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 0 & 0 & -1 & 1 & -1 & 1 \end{pmatrix} c,$$

where $c=\delta_x/\delta_t$ is the particle velocity, δ_x and δ_t are the lattice spacing and time step, respectively. $c=1$ is set in the following; therefore, the relevant quantities do not contain c.

The evolution equation of existing D3Q13 MRT LB models is

$$f_i(\boldsymbol{x}+\boldsymbol{c}_i\delta_t,t+\delta_t)-f_i(\boldsymbol{x},t)=-\sum_{i=0}^{12}\boldsymbol{\Lambda}_{ij}(f_j(\boldsymbol{x},t)-f_j^{(eq)}(\boldsymbol{x},t)),\quad i=0-12, \tag{2.1}$$

where $f_i(\boldsymbol{x},t)$ is the distribution function corresponding to velocity $\boldsymbol{c}_i$, $f_i^{(eq)}(\boldsymbol{x},t)$ is the equilibrium distribution function and $\boldsymbol{\Lambda}_{ij}$ is the element of a 13×13 collision matrix $\boldsymbol{\Lambda}$. The evolution equation can also be written in a vector form:

$$|f(\boldsymbol{x}+\boldsymbol{c}_i\delta_t,t+\delta_t)\rangle-|f(\boldsymbol{x},t)\rangle=-\boldsymbol{\Lambda}(|f(\boldsymbol{x},t)\rangle-|f^{(eq)}(\boldsymbol{x},t)\rangle), \tag{2.2}$$

where

$$|f(\boldsymbol{x},t)\rangle=(f_0(\boldsymbol{x},t),f_1(\boldsymbol{x},t),\cdots,f_{12}(\boldsymbol{x},t))',\quad |f^{(eq)}(\boldsymbol{x},t)\rangle=(f_0^{(eq)}(\boldsymbol{x},t),f_1^{(eq)}(\boldsymbol{x},t),\cdots,f_{12}^{(eq)}(\boldsymbol{x},t))',$$
$$|f(\boldsymbol{x}+\boldsymbol{c}_i\delta_t,t+\delta_t)\rangle=(f(\boldsymbol{x},t+\delta_t),f(\boldsymbol{x}+\boldsymbol{c}_1\delta_t,t+\delta_t),f(\boldsymbol{x}+\boldsymbol{c}_{12}\delta_t,t+\delta_t))',$$

and the superscript $'$ represents the transpose operator.

The transformation matrix of the existing D3Q13 MRT LB models [14] is

$$
\boldsymbol{T}=\begin{pmatrix}
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1\\
0 & 1 & 1 & 1 & 1 & 0 & 0 & -1 & -1 & -1 & -1 & 0 & 0\\
0 & 1 & -1 & 0 & 0 & 1 & 1 & -1 & 1 & 0 & 0 & -1 & -1\\
0 & 0 & 0 & 1 & -1 & 1 & -1 & 0 & 0 & -1 & 1 & -1 & 1\\
-12 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1\\
0 & 1 & 1 & 1 & 1 & -2 & -2 & 1 & 1 & 1 & 1 & -2 & -2\\
0 & 1 & 1 & -1 & -1 & 0 & 0 & 1 & 1 & -1 & -1 & 0 & 0\\
0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & -1\\
0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0\\
0 & 1 & 1 & -1 & -1 & 0 & 0 & -1 & -1 & 1 & 1 & 0 & 0\\
0 & -1 & 1 & 0 & 0 & 1 & 1 & 1 & -1 & 0 & 0 & -1 & -1\\
0 & 0 & 0 & 1 & -1 & -1 & 1 & 0 & 0 & -1 & 1 & 1 & -1
\end{pmatrix}. \tag{2.3}
$$

Thirteen moments are defined by linear mapping $\boldsymbol{m}(\boldsymbol{x},t)=\boldsymbol{T}|f(\boldsymbol{x},t)\rangle$ and

$$\boldsymbol{m}(\boldsymbol{x},t)=(\rho,j_x,j_y,j_z,e,3S_{xx},S_{ww},S_{xy},S_{yz},S_{xz},h_x,h_y,h_z)', \tag{2.4}$$

where the components of the vector $\boldsymbol{m}(\boldsymbol{x},t)$ correspond to the density, the mass fluxes along x, y and z, a scalar related to the kinetic energy, five moments related to the viscous stress tensor, and three without obvious physical meaning, respectively.

Suppose that the collision matrix is defined as $\boldsymbol{\Lambda}=\boldsymbol{T}^{-1}\hat{\boldsymbol{\Lambda}}\boldsymbol{T}$ and

$$\hat{\boldsymbol{\Lambda}}=diag(\lambda_c,\lambda_c,\lambda_c,\lambda_c,\lambda_e,\lambda_v,\lambda_v,\lambda_v',\lambda_v',\lambda_v',\lambda_h,\lambda_h,\lambda_h), \tag{2.5}$$

where λ_c, λ_e, λ_v, λ_v' and λ_h are the relaxation parameters corresponding to the conserved moments and the moments related to the kinetic energy, the viscous stress tensor and other moments without physical meaning, respectively.

To recover the N-S equations, one can choose the equilibrium moments of the original D3Q13 MRT LB model as

$$\rho^{(eq)}=\rho, \quad j_x^{(eq)}=\rho u_x, \quad j_y^{(eq)}=\rho u_y, \quad j_z^{(eq)}=\rho u_z, \quad e^{(eq)}=\frac{3}{2}(13c_s^2-8)\rho+\frac{13}{2}\rho u^2, \tag{2.6a}$$

$$3S_{xx}^{(eq)}=\rho(3u_x^2-u^2), \quad S_{ww}^{(eq)}=\rho(u_y^2-u_z^2), \quad S_{xy}^{(eq)}=\rho u_x u_y, \quad S_{yz}^{(eq)}=\rho u_y u_z, \tag{2.6b}$$

$$S_{xz}^{(eq)}=\rho u_x u_z, \quad h_x^{(eq)}=0, \quad h_y^{(eq)}=0, \quad h_z^{(eq)}=0, \tag{2.6c}$$

where ρ is the density, u_x, u_y and u_z are the velocities in x, y and z directions in a Cartesian coordinate system, respectively, c_s is the speed of sound, u is the modulus of velocity, that is,

$$|\boldsymbol{u}|=|(u_x,u_y,u_z)|=\sqrt{u_x^2+u_y^2+u_z^2}.$$

Therefore, the evolution equation can be written as

$$|f(\boldsymbol{x}+\boldsymbol{c}_i\delta_t,t+\delta_t)\rangle=\boldsymbol{T}^{-1}[\boldsymbol{m}(\boldsymbol{x},t)-\hat{\boldsymbol{\Lambda}}(\boldsymbol{m}(\boldsymbol{x},t)-\boldsymbol{m}^{(eq)}(\boldsymbol{x},t))], \tag{2.7}$$

where

$$\begin{aligned}&\boldsymbol{m}^{(eq)}(\boldsymbol{x},t)=\boldsymbol{T}|f^{(eq)}(\boldsymbol{x},t)\rangle\\ =&(\rho^{(eq)},j_x^{(eq)},j_y^{(eq)},j_z^{(eq)},e^{(eq)},3S_{xx}^{(eq)},S_{ww}^{(eq)},S_{xy}^{(eq)},S_{yz}^{(eq)},S_{xz}^{(eq)},h_x^{(eq)},h_y^{(eq)},h_z^{(eq)})'.\end{aligned}$$

Via the Chapman-Enskog (C-E) analysis, the compressible N-S equations can be recovered under the low Mach number limit (equivalent to the incompressible limit) as

$$\frac{\partial\rho}{\partial t}+\nabla\cdot(\rho\boldsymbol{u})=0, \tag{2.8a}$$

$$\frac{\partial(\rho\boldsymbol{u})}{\partial t}+\nabla\cdot(\rho\boldsymbol{u}\boldsymbol{u})=-c_s^2\nabla\rho+\nu\Delta(\rho\boldsymbol{u})+\Big(\frac{\nu}{3}+\zeta\Big)\nabla\nabla\cdot(\rho\boldsymbol{u}), \tag{2.8b}$$

with the shear and bulk kinematic viscosities

$$\nu=\frac{1}{4}\Big(\frac{1}{\lambda_\nu}-\frac{1}{2}\Big)=\frac{1}{2}\Big(\frac{1}{\lambda'_\nu}-\frac{1}{2}\Big),\quad \zeta=\Big(\frac{2}{3}-c_s^2\Big)\Big(\frac{1}{\lambda_e}-\frac{1}{2}\Big). \tag{2.9}$$

The density ρ, pressure p and velocity $\boldsymbol{u}$ can be obtained by

$$\rho=\sum_{i=0}^{12}f_i,\quad p=c_s^2\rho,\quad \boldsymbol{u}=\sum_{i=1}^{12}\boldsymbol{c}_if_i. \tag{2.10}$$

d'Humières also pointed out that the so-called "incompressible" D3Q13 MRT LB model can be obtained by using the following equilibrium moments [14]:

$$\rho^{(eq)}=\rho,\quad j_x^{(eq)}=\rho_0u_x,\quad j_y^{(eq)}=\rho_0u_y,\quad j_z^{(eq)}=\rho_0u_z, \tag{2.11a}$$

$$e^{(eq)}=\frac{3}{2}(13c_s^2-8)\rho+\frac{13}{2}\rho_0u^2,\quad 3S_{xx}^{(eq)}=\rho_0(3u_x^2-u^2),\quad S_{ww}^{(eq)}=\rho_0(u_y^2-u_z^2), \tag{2.11b}$$

$$S_{xy}^{(eq)}=\rho_0u_xu_y,\quad S_{yz}^{(eq)}=\rho_0u_yu_z,\quad S_{xz}^{(eq)}=\rho_0u_xu_z,\quad h_x^{(eq)}=0,\quad h_y^{(eq)}=0,\quad h_z^{(eq)}=0, \tag{2.11c}$$

where $\rho=\rho_0+\delta\rho$ is the density, ρ_0 is the average density, $\delta\rho$ is the density variation.

Via the C-E analysis, the macroscopic equations can be recovered under the low Mach number limit as

$$\frac{1}{c_s^2}\frac{\partial p}{\partial t}+\nabla\cdot\boldsymbol{u}=0, \tag{2.12a}$$

$$\frac{\partial\boldsymbol{u}}{\partial t}+\boldsymbol{u}\cdot\nabla\boldsymbol{u}=-\nabla p+\nu\Delta\boldsymbol{u}+\Big(\frac{\nu}{3}+\zeta\Big)\nabla(\nabla\cdot\boldsymbol{u}), \tag{2.12b}$$

with the shear and bulk kinematic viscosities determined by Eq. (2.9).

Clearly, in the case of steady flow, $\frac{\partial p}{\partial t}=0$, the condition for incompressible flow, $\nabla\cdot\boldsymbol{u}=0$, is satisfied exactly. The momentum equation becomes

$$\frac{\partial \boldsymbol{u}}{\partial t}+\boldsymbol{u}\cdot\nabla\boldsymbol{u}=-\nabla p+\nu\Delta\boldsymbol{u},$$

which is exactly the momentum equation of incompressible N-S equations. As in the case of unsteady flow, we can rewrite Eq. (2.12a) in a dimensionless form as follows

$$\frac{1}{T}\frac{\partial p'}{\partial t'}+\frac{c_s}{L}\nabla'\cdot\boldsymbol{u}'=0, \tag{2.13}$$

where $p'=p/c_s^2$, $t'=t/T$, $\nabla'=L\nabla$, $\boldsymbol{u}'=\boldsymbol{u}/c_s$, L and T are the characteristic length and time of the macroscopic flows, respectively. From Eq. (2.13), one can see that under an additional assumption $T\gg L/c_s$ the term $\frac{\partial p'}{\partial t'}$ can be negligible. Then, the macroscopic equations (2.12) can become the exact incompressible N-S equations.

For the "incompressible" D3Q13 MRT LB model, the density ρ, pressure p and velocity $\boldsymbol{u}$ can be obtained by

$$\rho=\sum_{i=0}^{12}f_i,\quad p=\frac{c_s^2\rho}{\rho_0},\quad \rho_0\boldsymbol{u}=\sum_{i=0}^{12}\boldsymbol{c}_i f_i. \tag{2.14}$$

3 The current D3Q13 MRT LB model and the methods to construct this model

3.1 The current D3Q13 MRT LB model

Except for the equilibrium moments, the discrete velocities, the evolution equation, the collision matrix, the transformation matrix, and the moments of the current D3Q13 MRT LB model are the same as those of the previous D3Q13 MRT LB models. To recover the exact incompressible N-S equations, we choose the equilibrium moments of the current D3Q13 MRT LB model as

$$\rho^{(eq)}=\rho_0,\quad j_x^{(eq)}=u_x,\quad j_y^{(eq)}=u_y,\quad j_z^{(eq)}=u_z,\quad e^{(eq)}=\frac{13u^2}{2}+\frac{39p}{2}-12\rho_0, \tag{3.1a}$$

$$3S_{xx}^{(eq)}=3u_x^2-u^2,\quad S_{ww}^{(eq)}=u_y^2-u_z^2,\quad S_{xy}^{(eq)}=u_xu_y,\quad S_{yz}^{(eq)}=u_yu_z, \tag{3.1b}$$

$$S_{xz}^{(eq)}=u_xu_z,\quad h_x^{(eq)}=0,\quad h_y^{(eq)}=0,\quad h_z^{(eq)}=0, \tag{3.1c}$$

where ρ_0 is the constant density and p is the pressure of fluid.

Via the C-E analysis, the exact incompressible N-S equations can be recovered only under the low Mach number assumption as

$$\nabla\cdot\boldsymbol{u}=0, \tag{3.2a}$$

$$\frac{\partial \boldsymbol{u}}{\partial t}+\boldsymbol{u}\cdot\nabla\boldsymbol{u}=-\nabla p+\nu\nabla^2\boldsymbol{u}, \tag{3.2b}$$

with the shear kinematic viscosity

$$\nu=\frac{1}{4}\Big(\frac{1}{\lambda_\nu}-\frac{1}{2}\Big)=\frac{1}{2}\Big(\frac{1}{\lambda'_\nu}-\frac{1}{2}\Big). \tag{3.3}$$

We can see that under the low Mach number assumption, the recovered macroscopic equations of the current D3Q13 MRT LB model are the exact incompressible N-S equations, while the recovered macroscopic equations of the previous D3Q13 MRT LB models are not. This is one of the main differences between the current D3Q13 MRT LB model and the previous D3Q13 MRT LB models.

The density ρ_0, pressure p and velocity $\boldsymbol{u}$ can be obtained by

$$\rho_0=\sum_{i=0}^{12}f_i,\quad p=\frac{2}{3}\Big(\sum_{i=1}^{12}f_i-\frac{|\boldsymbol{u}|^2}{2}\Big),\quad \boldsymbol{u}=\sum_{i=1}^{12}\boldsymbol{c}_i f_i. \tag{3.4}$$

From the above equation, we can see that the density of the fluid is always equal to ρ_0, so the density need not be computed in the simulation and is usually set to be a constant. We can also see that the pressure is decoupled with the density. This is the other main difference between the current D3Q13 MRT LB model and the previous D3Q13 MRT LB models.

3.2 The method to design the equilibrium moments of the current D3Q13 MRT LB model

The equilibrium moments $\boldsymbol{m}^{(eq)}$ of the MRT LB model and the equilibrium distribution functions (EDFs) $|f^{(eq)}(\boldsymbol{x},t)\rangle$ of the corresponding single-relaxation-time (SRT) LB model (also called LBGK model) generally have the relationship $\boldsymbol{m}^{(eq)}=\boldsymbol{T}|f^{(eq)}(\boldsymbol{x},t)\rangle$. Therefore, we first use the inverse of the transformation matrix of existing D3Q13 MRT LB models to multiply the equilibrium moments of the original D3Q13 MRT LB model and get

$$|f^{(eq)}(\boldsymbol{x},t)\rangle=\begin{pmatrix}
-(\rho(3c_s^2+u^2-2))/2\\
(\rho(c_s^2+u_x^2+2u_xu_y+u_x+u_y^2+u_y-u_z^2))/8\\
(\rho(c_s^2+u_x^2-2u_xu_y+u_x+u_y^2-u_y-u_z^2))/8\\
(\rho(c_s^2+u_x^2+2u_xu_z+u_x-u_y^2+u_z^2+u_z))/8\\
(\rho(c_s^2+u_x^2-2u_xu_z+u_x-u_y^2+u_z^2-u_z))/8\\
(\rho(c_s^2+u^2-2u_x^2+u_y+u_z+2u_yu_z))/8\\
(\rho(c_s^2+u^2-2u_x^2+u_y-u_z-2u_yu_z))/8\\
(\rho(c_s^2+u_x^2+2u_xu_y-u_x+u_y^2-u_y-u_z^2))/8\\
(\rho(c_s^2+u_x^2-2u_xu_y-u_x+u_y^2+u_y-u_z^2))/8\\
(\rho(c_s^2+u_x^2+2u_xu_z-u_x-u_y^2+u_z^2-u_z))/8\\
(\rho(c_s^2+u_x^2-2u_xu_z-u_x-u_y^2+u_z^2+u_z))/8\\
-(\rho(-c_s^2-u^2+2u_x^2+u_y+u_z-2u_yu_z))/8\\
-(\rho(-c_s^2-u^2+2u_x^2+u_y-u_z+2u_yu_z))/8
\end{pmatrix}. \tag{3.5}$$

We replace the density ρ multiplied by the velocities in the above EDFs with the average density ρ_0 and get the following EDFs

$$|f^{(eq)}(\boldsymbol{x},t)\rangle=\begin{pmatrix} -(\rho(3c_s^2-2)+\rho_0u^2)/2 \\ (\rho c_s^2+\rho_0(u_x^2+2u_xu_y+u_x+u_y^2+u_y-u_z^2))/8 \\ (\rho c_s^2+\rho_0(u_x^2-2u_xu_y+u_x+u_y^2-u_y-u_z^2))/8 \\ (\rho c_s^2+\rho_0(u_x^2+2u_xu_z+u_x-u_y^2+u_z^2+u_z))/8 \\ (\rho c_s^2+\rho_0(u_x^2-2u_xu_z+u_x-u_y^2+u_z^2-u_z))/8 \\ (\rho c_s^2+\rho_0(u^2-2u_x^2+u_y+u_z+2u_yu_z))/8 \\ (\rho c_s^2+\rho_0(u^2-2u_x^2+u_y-u_z-2u_yu_z))/8 \\ (\rho c_s^2+\rho_0(u_x^2+2u_xu_y-u_x+u_y^2-u_y-u_z^2))/8 \\ (\rho c_s^2+\rho_0(u_x^2-2u_xu_y-u_x+u_y^2+u_y-u_z^2))/8 \\ (\rho c_s^2+\rho_0(u_x^2+2u_xu_z-u_x-u_y^2+u_z^2-u_z))/8 \\ (\rho c_s^2+\rho_0(u_x^2-2u_xu_z-u_x-u_y^2+u_z^2+u_z))/8 \\ (\rho c_s^2+\rho_0(u^2-2u_x^2-u_y-u_z+2u_yu_z))/8 \\ (\rho c_s^2+\rho_0(u^2-2u_x^2-u_y+u_z-2u_yu_z))/8 \end{pmatrix}. \tag{3.6}$$

If we use the transformation matrix of the existing D3Q13 MRT LB models to multiply the above EDFs, we get the equilibrium moments of the "incompressible" D3Q13 MRT LB model. Then we replace ρ_0 multiplied by the velocities with 1 and replace the term $c_s^2\rho$ in the EDFs from $f_1^{(eq)}(\boldsymbol{x},t)$ to $f_{12}^{(eq)}(\boldsymbol{x},t)$ with the pressure p, and define the EDF in the direction $\boldsymbol{c}_0$ as ρ_0 minus the sum of EDFs from $f_1^{(eq)}(\boldsymbol{x},t)$ to $f_{12}^{(eq)}(\boldsymbol{x},t)$, then we get the following EDFs

$$|f^{(eq)}(\boldsymbol{x},t)\rangle=\begin{pmatrix} -(3/2)p+\rho_0-u^2/2 \\ (p+(u_x^2+2u_xu_y+u_x+u_y^2+u_y-u_z^2))/8 \\ (p+(u_x^2-2u_xu_y+u_x+u_y^2-u_y-u_z^2))/8 \\ (p+(u_x^2+2u_xu_z+u_x-u_y^2+u_z^2+u_z))/8 \\ (p+(u_x^2-2u_xu_z+u_x-u_y^2+u_z^2-u_z))/8 \\ (p+(u^2-2u_x^2+u_y+u_z+2u_yu_z))/8 \\ (p+(u^2-2u_x^2+u_y-u_z-2u_yu_z))/8 \\ (p+(u_x^2+2u_xu_y-u_x+u_y^2-u_y-u_z^2))/8 \\ (p+(u_x^2-2u_xu_y-u_x+u_y^2+u_y-u_z^2))/8 \\ (p+(u_x^2+2u_xu_z-u_x-u_y^2+u_z^2-u_z))/8 \\ (p+(u_x^2-2u_xu_z-u_x-u_y^2+u_z^2+u_z))/8 \\ (p+(u^2-2u_x^2-u_y-u_z+2u_yu_z))/8 \\ (p+(u^2-2u_x^2-u_y+u_z-2u_yu_z))/8 \end{pmatrix}. \tag{3.7}$$

If we use the transformation matrix of the existing D3Q13 MRT LB models to multiply the above EDFs, we will get the equilibrium moments of the current D3Q13 MRT LB model.

3.3 The method to derive the formula to compute the pressure in the current D3Q13 MRT LB model

The Appendix presents the derivation of macroscopic equations of the current D3Q13 MRT LB model. From Eq. (A.1), Eq. (A.6), and Eq. (A.13a), we know that $e^{(1)}$ is of order $\mathcal{O}(\delta_t+M^2)$. As $\boldsymbol{m}=\sum_{n=0}^{\infty}\varepsilon^n\boldsymbol{m}^{(n)}$, we have $e=e^{(0)}+\varepsilon e^{(1)}+\cdots$. From Eq. (A.4a), we know that $e=e^{(0)}=e^{(eq)}$ with the error of $\mathcal{O}(\delta_t{}^2+\delta_t M^2)$. From the definition of moments, we obtain

$$\sum_{i=1}^{12} f_i-12f_0=\frac{13}{2}|\boldsymbol{u}|^2+\frac{39}{2}p-12\rho_0.$$

By using $\rho_0=\sum_{i=0}^{12} f_i$, we have

$$p=\frac{2}{3}\Big(\sum_{i=1}^{12} f_i-\frac{|\boldsymbol{u}|^2}{2}\Big).$$

4 Numerical results and discussion

In this section, to verify the effectiveness of the current D3Q13 MRT LB model, we simulate three benchmark flow problems, i.e., the 3D Poiseuille flow, 3D pulsatile flow, and 3D lid-driven cavity flow. The relaxation parameters in the collision matrix are set as $\lambda_c=\lambda_e=1.0$, $\lambda_h=1.8$. We also present the simulation results of the "incompressible" D3Q13 MRT LB model and the D3Q19 MRT LB model [7]. For the "incompressible" D3Q13 MRT LB model, the values of relaxation parameters in the collision matrix are the same as those in the current D3Q13 MRT LB model. For the D3Q19 MRT LB model, the relaxation parameters in the collision matrix are set as $\lambda_c=1.0$, $\lambda_e=1.19$, $\lambda_\varepsilon=\lambda_\pi=1.0$, $\lambda_q=1.2$, $\lambda_m=0.98$. For the above three MRT LB models, the relaxation parameters related to the kinematic viscosity are determined by the corresponding equations of the three MRT LB models. For example, in the current D3Q13 MRT LB model, λ_ν and λ'_ν are determined by Eq. (3.3). It should be noted that, in the following simulation, c is not always equal to 1. The treatment can be referred to [6]. In addition, the non-equilibrium extrapolation scheme [21] is used to treat the boundary condition.

The time-stepping method for the current D3Q13 MRT LB model, the "incompressible" D3Q13 MRT LB model, and the D3Q19 MRT LB model in the simulation of 3D Poiseuille flow, 3D pulsatile flow, and 3D lid-driven cavity flow is summarized as follows:

(1). Initialization: The values of parameters such as λ_ν, δt, and δx are first set. Then, the pressure and velocity are initialized based on the flows. Then, the initial equilibrium moments and moments are computed from Eq. (3.1). The initial distribution functions are determined by $|f(\boldsymbol{x},t)\rangle = \boldsymbol{T}^{-1}\boldsymbol{m}(\boldsymbol{x},t)$, where $\boldsymbol{T}$ is the transformation matrix as shown in Eq. (2.3).

(2). Collision: Firstly, we use $\boldsymbol{m}(\boldsymbol{x},t)=\boldsymbol{T}|f(\boldsymbol{x},t)\rangle$ to obtain the moments $\boldsymbol{m}(\boldsymbol{x},t)$. Then, we use Eq. (3.1) to compute the $\boldsymbol{m}^{(eq)}(\boldsymbol{x},t)$ based on the known pressure and velocity. Then, we compute $|f^*(\boldsymbol{x},t)\rangle=\boldsymbol{T}^{-1}[\boldsymbol{m}(\boldsymbol{x},t)-\hat{\boldsymbol{\Lambda}}(\boldsymbol{m}(\boldsymbol{x},t)-\boldsymbol{m}^{(eq)}(\boldsymbol{x},t))]$ based on the right-hand side of Eq. (2.7), where $|f^*(\boldsymbol{x},t)\rangle$ is the distribution function after a collision.

(3). Propagation: The propagation step is conducted based on Eq. (2.7), that is, $|f(\boldsymbol{x}+\boldsymbol{c}_i\delta t,t+\delta t)\rangle=|f^*(\boldsymbol{x},t)\rangle$. Suppose $f_i^*(\boldsymbol{x},t)$ $(0\le i\le 12)$ is one component of the vector $|f^*(\boldsymbol{x},t)\rangle$. In the propagation step, it is assigned to $f_i(\boldsymbol{x}+\boldsymbol{c}_i\delta t,t+\delta t)$, which is one component of the vector $|f(\boldsymbol{x}+\boldsymbol{c}_i\delta t,t+\delta t)\rangle$.

(4). Boundary condition treatment: The pressure and velocity on the boundary nodes at $t+\delta t$ are determined according to the boundary condition of the flows. The distribution functions on the boundary nodes at $t+\delta t$ are updated using the non-equilibrium extrapolation scheme [21].

(5). Calculation of macroscopic quantities: The pressure and velocity at $t+\delta t$ are computed from Eq. (3.4) with the distribution functions $f_i(\boldsymbol{x},t+\delta t)$ in the internal nodes of the computational domain.

(6). Go to step (2) if the error precision is not satisfied.

4.1 3D Poiseuille flow

The flow domain of three-dimensional Poiseuille flow and the boundary conditions are shown in Fig. 1. The flow region is defined in a square duct: $0\le x\le l$, $-a\le y\le a$ and $-b\le z\le b$, where $l=2$, $a=b=0.5$. If the boundary conditions are

$$\mathbf{u}(x,\pm a,z,t)=\mathbf{u}(x,y,\pm b,t)=0, \tag{4.1a}$$

$$p(0,y,z,t)=p_{in},\quad p(l,y,z,t)=p_{out}, \tag{4.1b}$$

where p_{in} and p_{out} are the pressure at the inlet and outlet of the channel, the flow governed by the incompressible N-S equations has a steady analytical solution [22],

$$u_x(y,z,t)=\frac{16a^2}{\rho_0\nu\pi^3}\left(-\frac{dp}{dx}\right)\sum_{i=1,3,5,\cdots,}^{\infty}(-1)^{(i-1)/2}\left[1-\frac{\cosh(i\pi z/2a)}{\cosh(i\pi b/2a)}\right]\times\frac{\cos(i\pi y/2a)}{i^3}, \tag{4.2a}$$

$$u_y(x,y,z,t)=u_z(x,y,z,t)=0, \tag{4.2b}$$

$$p(x,t)=p_{in}+\frac{dp}{dx}x, \tag{4.2c}$$

where $dp/dx=(p_{out}-p_{in})/l$ is the pressure gradient in the channel, ν is the kinematic viscosity of fluid.

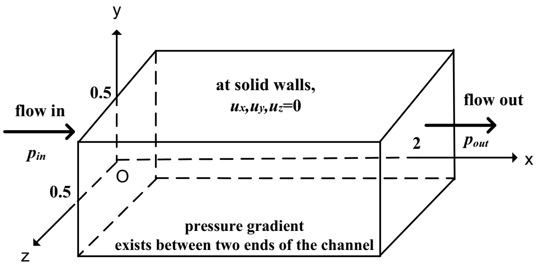


Figure 1: Simulation configurations of three-dimensional Poiseuille flow.

In the simulation, the initial condition is

$$\mathbf{u}(x,y,z,0)=0, \quad p(0<x<l,\ y,\ z,\ 0)=\frac{p_{in}+p_{out}}{2}, \tag{4.3}$$

where $p_{out}=1.0$, $p_{in}=p_{out}+\Delta p$ and Δp is set to be different values in the simulation. The criterion for the steady state is,

$$\frac{\sum_i |u_x(\boldsymbol{x}_i,t+100\delta t)-u_x(\boldsymbol{x}_i,t)|}{\sum_i |u_x(\boldsymbol{x}_i,t+100\delta_t)|} \leq 6.0\times 10^{-14}, \tag{4.4}$$

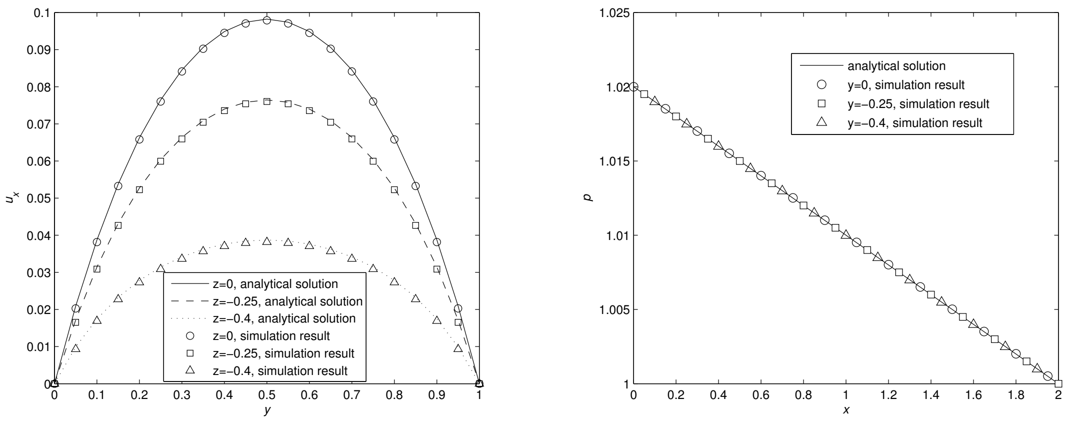
where $\sum_i$ denotes the summation over every grid point.

First, simulations are carried out for $\nu=0.0075$ and $\Delta p=0.02$ under the grid $N_x\times N_y\times N_z=41\times 21\times 21$ with the current D3Q13 MRT LB model. Fig. 2 shows the distribution of horizontal velocities in the y direction for different locations of z at section $x=1$ and the distribution of pressure in the x direction for different locations of y at section $z=0$. It can be seen from Fig. 2 that the numerical results are in excellent agreement with the analytical solutions.

We also conduct simulations for various values of Δp with the current D3Q13 MRT LB model, the "incompressible" D3Q13 MRT LB model, and the D3Q19 MRT LB model. The global relative error in the velocity field (GRE_u) between the numerical solution and the analytical solution is defined as

$$GRE_u=\frac{\sqrt{\sum_i \left[(u_{xn}-u_{xa})^2+(u_{yn}-u_{ya})^2+(u_{zn}-u_{za})^2\right]}}{\sqrt{\sum_i \left[u_{xa}^2+u_{ya}^2+u_{za}^2\right]}}, \tag{4.5}$$

where u_{xn}, u_{yn}, u_{zn} and u_{xa}, u_{ya}, u_{za} denote the velocity components of numerical and analytical solutions, respectively, and the summation of i is over every grid point. The GRE_u for different values of Δp and different MRT LB models are displayed in Table 1. From Table 1, we can see that the GRE_u of the current D3Q13 MRT LB model is the same as that



(a) (b)

Figure 2: (a) shows the horizontal velocity profiles for Poiseuille flow at the section $x=1$, while (b) shows the pressure profiles at the section $z=0$. The simulation results come from the current D3Q13 MRT LB model.

of the "incompressible" D3Q13 MRT LB model under each value of Δp. This shows that the current D3Q13 MRT LB model and the "incompressible" D3Q13 MRT LB model have the same accuracy in simulating steady 3D Poiseuille flow. We can also see that the GRE_u of the D3Q19 MRT LB model is the smallest under each value of Δp compared with the two D3Q13 MRT LB models. This may be because the D3Q19 MRT LB model has more discrete velocity directions than the two D3Q13 MRT LB models. As the lattice Boltzmann evolution equation can be treated as an approximation to the Boltzmann equation, more discrete velocity directions may lead to higher numerical accuracy.

Table 1: The variation of GRE_u with the pressure difference Δp for Poiseuille flow by using different MRT LB models.

Δp	GRE_u		
	current D3Q13 MRT model	"incompressible" D3Q13 MRT model	D3Q19 MRT model
0.001	8.4337×10^{-3}	8.4337×10^{-3}	4.2196×10^{-3}
0.005	8.4378×10^{-3}	8.4378×10^{-3}	4.2160×10^{-3}
0.01	8.4508×10^{-3}	8.4508×10^{-3}	4.2043×10^{-3}
0.015	8.4733×10^{-3}	8.4733×10^{-3}	4.1837×10^{-3}
0.02	8.5066×10^{-3}	8.5066×10^{-3}	4.1528×10^{-3}
0.025	8.5518×10^{-3}	8.5518×10^{-3}	4.1100×10^{-3}
0.03	8.6106×10^{-3}	8.6106×10^{-3}	4.0538×10^{-3}
0.035	8.6844×10^{-3}	8.6844×10^{-3}	3.9828×10^{-3}
0.04	8.7748×10^{-3}	8.7748×10^{-3}	3.8959×10^{-3}
0.045	8.8832×10^{-3}	8.8832×10^{-3}	3.7924×10^{-3}
0.05	9.0109×10^{-3}	9.0109×10^{-3}	3.6721×10^{-3}

4.2 3D Pulsatile flow

In this section, the 3D pulsatile flow is used to assess the current D3Q13 MRT LB model for the unsteady incompressible flows. The flow domain and the initial and boundary conditions of the 3D pulsatile flow are the same as those used for the 3D Poiseuille flow. The difference between the pulsatile flow and the Poiseuille flow is that the former's pressure gradient changes periodically. The pressure gradient is $\frac{dp}{dx}=\frac{\Delta p}{l}\cos(\omega t)$, where $\Delta p = p_{out} - p_{in}$ is the the maximal pressure difference between the two ends of the channel, l is the length of the channel, and ω is the frequency.

The analytical solution of 3D pulsatile flow is [23]:

$$\begin{aligned} u_x(y,z,t)=&\mathrm{Re}\left\{\frac{b^2\Delta p}{\eta^2\mu l}e^{-i\pi/2}\left\{1-2\sum_{n=0}^{\infty}\frac{(-1)^n}{p_n}\left[\frac{\cosh(\gamma_n y/b)\cos(p_n z/b)}{\cosh(\gamma_n a/b)}\right.\right.\right.\\ &\left.\left.\left.+\frac{\cosh(\sigma_n z/b)\cos(q_n y/b)}{\cosh(\sigma_n)}\right]\right\}e^{i\omega t}\right\}, \end{aligned} \tag{4.6}$$

where

$$\gamma_n=\sqrt{p_n^2+i\eta^2}, \qquad \sigma_n=\sqrt{q_n^2+i\eta^2}, \tag{4.7a}$$

$$p_n=\frac{2n+1}{2}\pi, \qquad q_n=\frac{2n+1}{2}\pi\frac{b}{a}, \tag{4.7b}$$

$\eta=b\sqrt{\omega/\nu}$ is the Womersley number, $\mu=\rho_0\nu$ is the dynamic viscosity of fluid.

The simulation parameters are set as follows. The grid size is $Nx\times Ny\times Nz=41\times 21\times 21$, the period of the changing pressure is $T=100$ ($\omega=2\pi/T$), the pressure at the outlet (p_{out}) is set to be 1.0. $\delta t=0.05$ is fixed in the simulation so that one period contains an integral step. The computation begins with $t=0$. After the velocity field satisfies the following convergence criterion,

$$\frac{\sum_i |u_x(\boldsymbol{x}_i,t+T)-u_x(\boldsymbol{x}_i,t)|}{\sum_i |u_x(\boldsymbol{x}_i,t+T)|}\le 6.0\times 10^{-14}, \tag{4.8}$$

we make the computation iterate more steps to obtain the velocity at different times, which is the ultimate numerical solution. Here, $|\cdot|$ denotes the absolute value operator, and the summation of i is over every grid point.

First, we conduct simulations to get the velocity profiles across the channel at different times with the current D3Q13 MRT LB model. Fig. 3 shows the velocity profiles in the lines $x=1$, $z=0$ at four different times: $t=T/4$, $T/2$, $3T/4$ and T under $\Delta p=0.02$. The fluid viscosity is set to be $\nu=0.0075$, and the corresponding Womersley number of the flow is $\eta=1.4472$. The velocity has been normalized with $U_{\max}$, which is the maximal horizontal velocity of Poiseuille flow with the pressure gradient $-\Delta p/l$. From the above figure, it can be seen that the numerical solution accords well with the analytical solution.

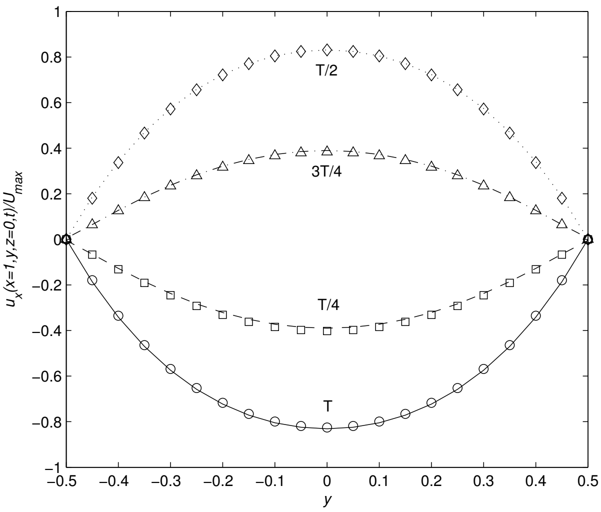


Figure 3: The variation of u_x with y for pulsatile flow at the location $x=1$, $z=0$, $\Delta p=0.02$, $\nu=0.0075$ and $\eta=1.4472$. lines: analytical solutions, symbols: numerical solutions of the current D3Q13 MRT LB model.

Secondly, we conduct simulations for different values of Δp varying from 0.001 to 0.05 with the current D3Q13 MRT LB model, the "incompressible" D3Q13 MRT LB model, and the D3Q19 MRT LB model. The GRE_u for different values of Δp and different MRT LB models are listed in Table 2. From Table 2, we can see that the GRE_u of the three MRT LB models for 3D pulsatile flow is larger than that for 3D Poiseuille flow. This may be because the pressure difference is a constant in the 3D Poiseuille flow, while

Table 2: The variation of GRE_u with the maximal pressure difference Δp for pulsatile flow at $t=T$ by using different MRT LB models, $\nu=0.0075$, $1/\lambda_\nu=1.1$.

Δp	GRE_u		
	current D3Q13 MRT model	"incompressible" D3Q13 MRT model	D3Q19 MRT model
0.001	1.0304×10^{-2}	1.1412×10^{-2}	5.7481×10^{-3}
0.005	1.0700×10^{-2}	1.2422×10^{-2}	6.3609×10^{-3}
0.01	1.1112×10^{-2}	1.3731×10^{-2}	7.0284×10^{-3}
0.015	1.1422×10^{-2}	1.5079×10^{-2}	7.5777×10^{-3}
0.02	1.1627×10^{-2}	1.6455×10^{-2}	8.0039×10^{-3}
0.025	1.1723×10^{-2}	1.7850×10^{-2}	8.3046×10^{-3}
0.03	1.1711×10^{-2}	1.9261×10^{-2}	8.4785×10^{-3}
0.035	1.1593×10^{-2}	2.0682×10^{-2}	8.5259×10^{-3}
0.04	1.1374×10^{-2}	2.2113×10^{-2}	8.4477×10^{-3}
0.045	1.1060×10^{-2}	2.3550×10^{-2}	8.2462×10^{-3}
0.05	1.0663×10^{-2}	2.4993×10^{-2}	7.9260×10^{-3}

in the pulsatile flow, the pressure difference is a periodic function of time. The flow becomes unsteady, which produces additional numerical errors for the three MRT LB models. From Table 2, we can also see that the GRE_u of the D3Q19 MRT LB model is still the smallest under each value of Δp compared with the case of 3D Poiseuille flow. However, the GRE_u of the current D3Q13 MRT LB model is always smaller than that of the "incompressible" D3Q13 MRT model. This is because the compressibility error of the second kind is eliminated in the current D3Q13 MRT LB model.

4.3 3D lid-driven cavity flow

In this section, the 3D lid-driven cavity flow is simulated to test the accuracy of the current D3Q13 MRT LB model. Based on this flow, we also evaluate the stability and efficiency of the current D3Q13 MRT LB model and compare this model with the "incompressible" D3Q13 MRT LB model and D3Q19 MRT LB model.

The 3D lid-driven cavity flow is defined in a cubic box, as shown in Fig. 4. The cubic box is in the 3D Cartesian coordinate, and its domain is $[0,1]\times[0,1]\times[0,1]$. The flow in the cubic box is driven by the top sliding lid with a constant velocity $U_0=1.0$. To make the flow satisfy the low Mach number assumption in the computation, we set $c=10$. Then, we can use the current D3Q13 MRT LB model to simulate the 3D lid-driven cavity flow.

Ku et al. simulated the 3D lid-driven cavity flow with the pseudospectral method [24].

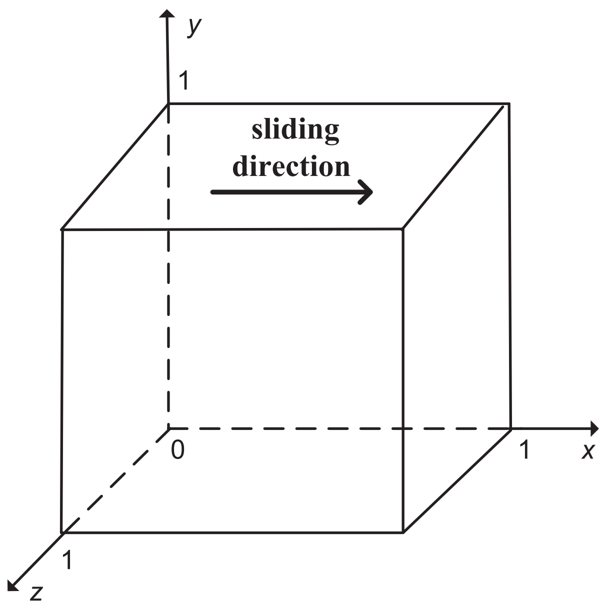
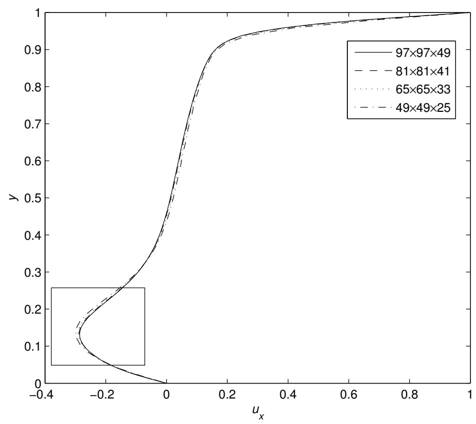
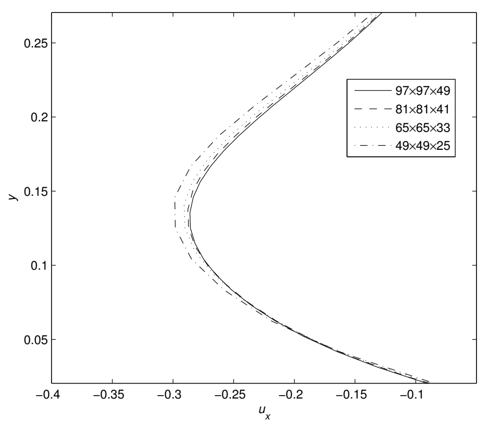


Figure 4: The schematic of three-dimensional lid-driven cavity flow.



(a)



(b)

Figure 5: The horizontal velocity distributions of the cavity flow at Re=1000 along the vertical center line at the section $z=0.5$ for four different grid sizes. The rectangle region in (a) is magnified and shown in (b).

Their results were also included to demonstrate the validity of the numerical methods involved [25–28]. Also, those published results validate the accuracy of the D3Q13 MRT LB model proposed in this paper. The initial velocities (except for the velocities on the sliding lid) and the pressure in the domain are all set to be zeros. At the section of $z=0.5$, the symmetric boundary condition is set, i.e.,

$$\frac{\partial u_x}{\partial z}=\frac{\partial u_y}{\partial z}=\frac{\partial p}{\partial z}=0, \quad u_z=0. \tag{4.9}$$

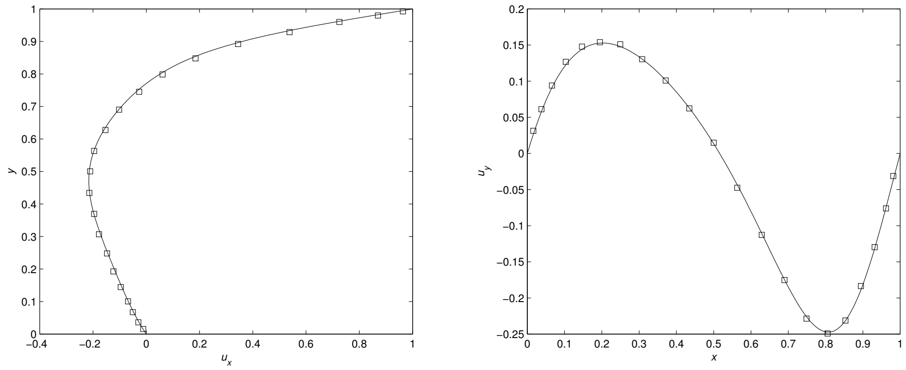
The convergence criterion for the velocity field is,

$$\frac{\sum_i\sum_k |u_k(\boldsymbol{x}_i,t+1000\delta t)-u_k(\boldsymbol{x}_i,t)|}{\sum_i\sum_k |u_k(\boldsymbol{x}_i,t+1000\delta t)|}\leq 6.0\times 10^{-14}, \tag{4.10}$$

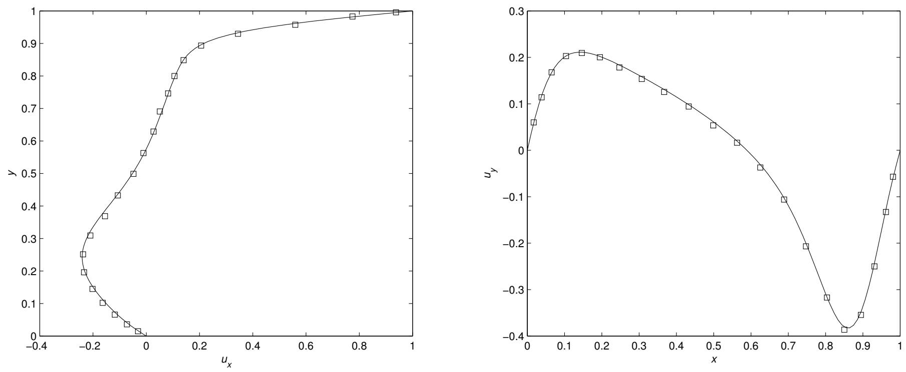
where the summations of i and k are carried out over all the grid points and the velocities in all directions.

First, the grid independence study of the simulation is carried out. The horizontal velocity distributions of the cavity flow at Re=1000 along the vertical center line at the section $z=0.5$ for four grids ($97\times 97\times 49$, $81\times 81\times 41$, $65\times 65\times 33$ and $49\times 49\times 25$) are shown in Fig. 5. From Fig. 5, it can be seen that the difference in the velocity distribution between the adjacent grid sizes becomes smaller and smaller as the grid size increases. In the following, the grid size $97\times 97\times 49$ is used to simulate cavity flow at different Reynolds numbers.

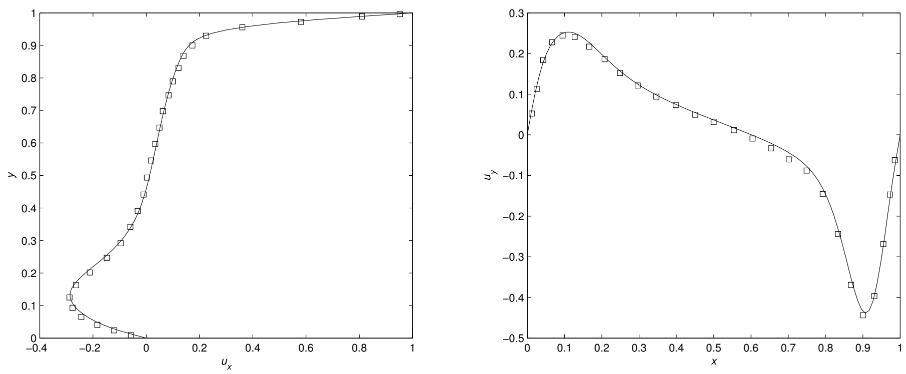
Secondly, the 3D lid-driven cavity flow at the Reynolds numbers 100, 400, and 1000 are simulated with the current D3Q13 MRT LB model under the grid $97\times 97\times 49$. The



(a) Re=100



(b) Re=400



(c) Re=1000

Figure 6: The velocity distribution in the vertical and horizontal center lines at the section $z=0.5$ for the cavity flow at different Reynolds numbers, □: the results of Ku et al., –: the simulation results by the current D3Q13 MRT LB model.

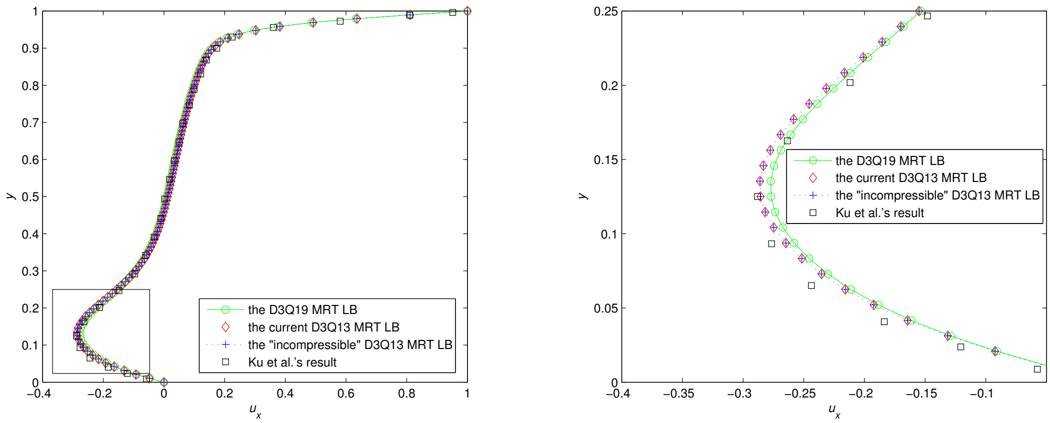
Table 3: The convergence or divergence of the simulation of the 3D cavity flow at some Reynolds number for the current D3Q13 MRT LB model, the "incompressible" D3Q13 MRT LB model, and the D3Q19 MRT LB model. The convergence criterion is that the computation does not blow up after 100000 iterations. In the table, the check mark means the simulation is convergent, while the cross means the simulation is divergent.

Re	Model type		
	current D3Q13 MRT model	"incompressible" D3Q13 MRT model	D3Q19 MRT model
100	✓	✓	✓
400	✓	✓	✓
1000	✓	✓	✓
1500	✓	✓	✓
7300	✓	✓	✓
7400	×	✓	✓
7500	×	✓	✓
7600	×	×	✓
18000	×	×	✓
18100	×	×	×

Reynolds number is defined as U_0L/ν, where $L=1.0$ is the side length of the cubic box. On this basis, the simulation results are compared with those of Ku et al. The comparison is shown in Fig. 6. It can be seen that the agreement between the results of the current D3Q13 MRT LB model and those of the Ku et al. method is excellent.

Thirdly, the D3Q13 MRT LB model is compared in terms of stability with the "incompressible" D3Q13 MRT LB model and the D3Q19 MRT LB model. The Reynolds number is increased constantly, and the convergence or divergence of the simulation of the 3D cavity flow at one Reynolds number under the grid $97\times97\times49$ is recorded. The record for the current D3Q13 MRT LB model, the "incompressible" D3Q13 MRT LB model, and the D3Q19 MRT LB model is shown in Table 3. From Table 3, we can see that the Reynolds number beyond which the simulation blew up (the critical Reynolds number) for the D3Q19 MRT LB model is much higher than those of the current D3Q13 MRT LB model and the "incompressible" D3Q13 MRT LB model. This may be attributed to the D3Q19 MRT LB model having more discrete velocity directions than the current D3Q13 MRT LB model and the "incompressible" D3Q13 MRT LB model. However, the critical Reynolds number of the current D3Q13 MRT LB model is almost as high as that of the "incompressible" D3Q13 MRT LB model. This may be because the current D3Q13 MRT LB model and the "incompressible" D3Q13 MRT LB model have the same discrete velocity directions.

Fourth, the accuracy of the current D3Q13 MRT LB model, the "incompressible" D3Q13 MRT LB model, and the D3Q19 MRT LB model are compared by simulating the cavity flow at Re=1000 under the grid $97\times97\times49$. All three MRT LB models can simulate the 3D cavity flow at Re=1000. As seen from Fig. 7(a), the horizontal velocity distributions of the cavity flow at Re=1000 along the vertical center line at the section $z=0.5$ for the three MRT LB models are all in excellent agreement with the Ku et al. result. From Fig. 7(b), it can be seen that the horizontal velocity distributions for the two D3Q13 MRT LB models are the same. Moreover, in some locations, the results of the two D3Q13 MRT



(a) (b)

Figure 7: The comparison of the horizontal velocity distributions of the cavity flow at Re=1000 along the vertical center line at the section $z=0.5$ for the current D3Q13 MRT LB model, the "incompressible" D3Q13 MRT LB model and the D3Q19 MRT LB model. The rectangle region in (a) is magnified and shown in (b).

LB models agree better with the Ku et al. result. The D3Q19 MRT LB model's result in some other locations agrees better with the Ku et al. result. It can be inferred that the accuracy of the current D3Q13 MRT LB model, the "incompressible" D3Q13 MRT LB model, and the D3Q19 MRT LB model are nearly the same when the three MRT LB models are used to simulate the 3D cavity flow under the critical Reynolds numbers.

Lastly, the computational efficiency of the current D3Q13 MRT LB model, the "incompressible" D3Q13 MRT LB model, and the D3Q19 MRT LB model are compared by simulating the cavity flow at Re=100, 400, and 1000 under the grid $97\times97\times49$. The computation is completed on a Tesla K40 graphic processing unit (GPU) card, and the code is compiled with the compute unified device architecture (CUDA). The computation time of the above three MRT LB models is listed in Table 4. From Table 4, we can see that the total and average run time of the current D3Q13 MRT LB model is very close to that of the "incompressible" D3Q13 MRT LB model. However, the total and average run time

Table 4: The computation time of the current D3Q13 MRT LB model (current D3Q13), the "incompressible" D3Q13 MRT LB model ("incompressible" D3Q13) and the D3Q19 MRT LB model (D3Q19) in simulating the 3D cavity flow at Re=100, 400, and 1000. "steps" denotes the number of steps that the computation needs to reach the convergence criterion. "time" denotes the total run time corresponding to the "steps." "ave time" denotes the average run time per 10^4 steps. The units of "time" and "ave time" are both seconds (s).

Re	current D3Q13			"incompressible" D3Q13			D3Q19		
	steps	time	ave time	steps	time	ave time	steps	time	ave time
100	4×10^4	59.67s	14.91s	4.0×10^4	57.77s	14.44s	4×10^4	98.07s	24.52s
400	9.3×10^4	138.64s	14.90s	9.3×10^4	134.06s	14.41s	9.3×10^4	227.87s	24.50s
1000	1.86×10^5	277.23s	14.90s	1.85×10^5	265.17s	14.33s	1.93×10^5	473.10s	24.51s

of the current D3Q13 MRT LB model are approximately 60% of those of the D3Q19 MRT model. This may be attributed to the fact that the current D3Q13 MRT LB model and the "incompressible" D3Q13 MRT LB model only have the 68% of the discrete velocity directions of the D3Q19 MRT LB model.

5 Conclusions

An alternative D3Q13 MRT LB model for incompressible Navier-Stokes equations is proposed in this study. The current D3Q13 MRT LB model can recover to the exact incompressible N-S equations only under the low Mach number assumption. This means the current D3Q13 MRT LB model can eliminate the compressibility error of the second kind associated with the solution of the incompressible N-S equations. Three benchmark flow problems, including 3D steady Poiseuille flow, 3D unsteady pulsatile flow, and 3D lid-driven cavity flow, are simulated to confirm the effectiveness of the current D3Q13 MRT LB model. The previous "incompressible" D3Q13 MRT LB model and the D3Q19 MRT LB model are also used to simulate the above three benchmark flow problems.

First, the three MRT LB models above simulate the 3D steady Poiseuille flow and 3D unsteady pulsatile flow under different (maximal) pressure differences. The GRE_u of the current D3Q13 MRT LB model is the same as that of the previous "incompressible" D3Q13 MRT LB model when the two models are used to solve the velocity field for the 3D steady Poiseuille flow under different pressure differences. However, the GRE_u of the current D3Q13 MRT LB model is always smaller than that of the previous "incompressible" D3Q13 MRT LB model when the two models are used to solve the velocity field for the 3D unsteady pulsatile flow under different maximal pressure differences. This improvement may be attributed to the fact that the compressibility error of the second kind is eliminated in the current D3Q13 MRT LB model. The GRE_u of the D3Q19 MRT LB model is the smallest among the three MRT LB models when they are used to solve the velocity field for the 3D steady Poiseuille flow and the 3D unsteady pulsatile flow under different (maximal) pressure differences. The reason may be that the D3Q19 MRT LB model has 46% more discrete velocity directions than the two D3Q13 MRT LB models.

The three MRT LB models also simulate the 3D lid-driven cavity flow at Re=100, 400, 1000, and higher Reynolds numbers. The D3Q19 MRT LB model can simulate the 3D lid-driven cavity flow at a much higher Reynolds number than the current D3Q13 MRT LB model and the "incompressible" D3Q13 MRT LB model. However, the current D3Q13 MRT LB model and the "incompressible" D3Q13 MRT LB model need much less runtime than the D3Q19 MRT LB model to complete the simulation of 3D lid-driven cavity flow at Re=100, 400, 1000. This difference is because the D3Q19 MRT LB model has more discrete velocity directions than the current D3Q13 MRT LB model and the "incompressible" D3Q13 MRT LB model. The previous "incompressible" D3Q13 MRT LB model can simulate the 3D lid-driven cavity flow at a slightly higher Reynolds number than the current D3Q13 MRT LB model. However, their runtime to simulate 3D lid-driven cavity flow at Re=100, 400, and 1000 is nearly identical.

It is established that for the MRT LB models, the relaxation parameters need to be adjusted according to the types of flow and the boundary treatment methods [29], so it is easier to set a smaller set of relaxation parameters for the MRT LB models. As the d-dimensional q-velocity MRT LB model generally has relaxation parameters comparable to q, the setting of the relaxation parameters for the current D3Q13 MRT LB model is more accessible than that for the D3Q15, D3Q19, and D3Q27 MRT LB models. More importantly, the current D3Q13 MRT LB model and the D3Q19 MRT LB model have nearly the same accuracy when the two models simulate the 3D cavity flow under one Reynolds number below the minimum of the critical Reynolds numbers of the two models. However, the computational efficiency of the current D3Q13 MRT LB model is much higher than the D3Q19 MRT LB model. Therefore, the current D3Q13 MRT LB model has a significant advantage over the D3Q19 MRT LB model in computational efficiency when the two models can simulate the flows with nearly the same accuracy.

Appendix

In this section, we perform the Chapman-Enskog analysis for the current D3Q13 MRT LB model to recover to the incompressible Navier-Stokes equations. We know that for an incompressible flow,

$$\mathcal{O}(\delta p)=\mathcal{O}(\delta\rho)=\mathcal{O}(M^2),\quad \mathcal{O}(\boldsymbol{u})=\mathcal{O}(M), \tag{A.1}$$

where M represents the Mach number, δp and $\delta\rho$ are the pressure and density fluctuations, respectively. We first introduce the Taylor series expansion and multi-scale expansions:

$$f_i(\boldsymbol{x}+\boldsymbol{c}_i\delta_t,t+\delta_t)=\sum_{n=0}^{\infty}\frac{\varepsilon^n}{n!}D_t^n f_i(\boldsymbol{x},t),\quad f_i=\sum_{n=0}^{\infty}\varepsilon^n f_i^{(n)},\quad \partial_t=\sum_{n=0}^{\infty}\varepsilon^n\partial_{t_n}, \tag{A.2}$$

where $\varepsilon=\delta_t$ and $D_t=\partial_t+\boldsymbol{c}_i\cdot\nabla=\partial_t+\boldsymbol{c}_{i\alpha}\cdot\partial_\alpha$. By applying Taylor series expansion and the above multi-scale expansions, we can rewrite the lattice Boltzmann equation (2.1) in different orders of ε as follows:

$$\mathcal{O}(\varepsilon^0):\quad f_i^{(0)}=f_i^{(eq)}, \tag{A.3a}$$

$$\mathcal{O}(\varepsilon^1):\quad D_{t_0}f_i^{(0)}=-\sum_{j=0}^{12}\Lambda_{ij}f_j^{(1)}, \tag{A.3b}$$

$$\mathcal{O}(\varepsilon^2):\quad \partial_{t_1}f_i^{(0)}+D_{t_0}\Big(f_i^{(1)}-\frac{1}{2}\sum_{j=0}^{12}\Lambda_{ij}f_j^{(1)}\Big)=-\sum_{j=0}^{12}\Lambda_{ij}f_j^{(2)}, \tag{A.3c}$$

where $D_{t_0}=\partial_{t_0}+\boldsymbol{c}_i\cdot\nabla$, and Eq. (A.3b) has been substituted into Eq. (A.3c). We can transform Eq. (A.3) into the moment space:

$$\boldsymbol{m}^{(0)}=\boldsymbol{m}^{(eq)}, \tag{A.4a}$$

$$(\partial_{t_0}\boldsymbol{I}+\hat{\boldsymbol{C}}_k\partial_k)\boldsymbol{m}^{(0)}=-\hat{\boldsymbol{\Lambda}}\boldsymbol{m}^{(1)}, \tag{A.4b}$$

$$\partial_{t_1}\boldsymbol{m}^{(0)}+(\partial_{t_0}\boldsymbol{I}+\hat{\boldsymbol{C}}_k\partial_k)\Big(\boldsymbol{I}-\frac{1}{2}\hat{\boldsymbol{\Lambda}}\Big)\boldsymbol{m}^{(1)}=-\hat{\boldsymbol{\Lambda}}\boldsymbol{m}^{(2)}, \tag{A.4c}$$

where $\hat{\boldsymbol{C}}_k=\boldsymbol{T}\boldsymbol{C}_k\boldsymbol{T}^{-1}$, $\boldsymbol{C}_k$ is a diagonal matrix with the k-th component of every discrete velocity $\boldsymbol{c}_i$ as the element, and

$$\boldsymbol{m}^{(n)}=\big(0,0,0,0,e^{(n)},3S_{xx}^{(n)},S_{\omega\omega}^{(n)},S_{xy}^{(n)},S_{yz}^{(n)},S_{xz}^{(n)},h_x^{(n)},h_y^{(n)},h_z^{(n)}\big)', \quad n=1,2. \tag{A.5}$$

Expanding Eq. (A.4b), we have

$$\begin{aligned}
&\partial_{t_0}\begin{bmatrix}\rho_0\\ u_x\\ u_y\\ u_z\\ 13u^2/2+39p/2-12\rho_0\\ 3u_x^2-u^2\\ u_y^2-u_z^2\\ u_xu_y\\ u_yu_z\\ u_xu_z\\ 0\\ 0\\ 0\end{bmatrix}
+\partial_x\begin{bmatrix}u_x\\ u_x^2+p\\ u_xu_y\\ u_xu_z\\ u_x\\ u_x\\ 0\\ u_y/2\\ 0\\ u_z/2\\ u_y^2-u_z^2\\ -u_xu_y\\ u_xu_z\end{bmatrix}
+\partial_y\begin{bmatrix}u_y\\ u_xu_y\\ u_y^2+p\\ u_yu_z\\ u_y\\ -u_y/2\\ u_y/2\\ u_x/2\\ u_z/2\\ 0\\ u_xu_y\\ u_z^2-u_x^2\\ -u_yu_z\end{bmatrix}\\
&+\partial_z\begin{bmatrix}u_z\\ u_xu_z\\ u_yu_z\\ u_z^2+p\\ u_y\\ -u_z/2\\ -u_z/2\\ 0\\ u_y/2\\ u_x/2\\ -u_xu_z\\ u_yu_z\\ u_x^2-u_y^2\end{bmatrix}
=-\begin{bmatrix}0\\ 0\\ 0\\ 0\\ -\lambda_e e^{(1)}\\ -\lambda_v 3S_{xx}^{(1)}\\ -\lambda_v S_{\omega\omega}^{(1)}\\ -\lambda_v' S_{xy}^{(1)}\\ -\lambda_v' S_{yz}^{(1)}\\ -\lambda_v' S_{xz}^{(1)}\\ -\lambda_h h_x^{(1)}\\ -\lambda_h h_y^{(1)}\\ -\lambda_h h_z^{(1)}\end{bmatrix}.
\end{aligned} \tag{A.6}$$

Taking the 1st to 4th rows of the above equations, we obtain

$$\partial_{t_0}\rho_0+\partial_x u_x+\partial_y u_y+\partial_z u_z=0. \tag{A.7}$$

and

$$\begin{aligned}
&\partial_{t_0}u_x+\partial_x(p+u_x^2)+\partial_y(u_xu_y)+\partial_z(u_xu_z)=0, &\text{(A.8a)}\\
&\partial_{t_0}u_y+\partial_x(u_xu_y)+\partial_y(p+u_y^2)+\partial_z(u_yu_z)=0, &\text{(A.8b)}\\
&\partial_{t_0}u_z+\partial_x(u_xu_z)+\partial_y(u_yu_z)+\partial_z(p+u_z^2)=0. &\text{(A.8c)}
\end{aligned}$$

As ρ_0 is a constant, Eq. (A.7) can lead to the continuity equation of the incompressible N-S equations.

From the expansion of Eq. (A.4c), we have

$$\begin{aligned}
&\partial_{t_1}u_x+\partial_x[(1-\lambda_\nu/2)S_{xx}^{(1)}+2(1-\lambda_e/2)e^{(1)}/39]+\partial_y[(1-\lambda_\nu'/2)S_{xy}^{(1)}]\\
&\quad+\partial_z[(1-\lambda_\nu'/2)S_{xz}^{(1)}]=0, &\text{(A.9a)}\\
&\partial_{t_1}u_y+\partial_x[(1-\lambda_\nu'/2)S_{xy}^{(1)}]+\partial_y[(1-\lambda_\nu/2)(S_{\omega\omega}^{(1)}-S_{xx}^{(1)})/2+2(1-\lambda_e/2)e^{(1)}/39]\\
&\quad+\partial_z[(1-\lambda_\nu'/2)S_{yz}^{(1)}]=0, &\text{(A.9b)}\\
&\partial_{t_1}u_z+\partial_x[(1-\lambda_\nu'/2)S_{xz}^{(1)}]+\partial_z[2(1-\lambda_e/2)e^{(1)}/39-(1-\lambda_\nu/2)(S_{\omega\omega}^{(1)}+S_{xx}^{(1)})/2]\\
&\quad+\partial_y[(1-\lambda_\nu'/2)S_{yz}^{(1)}]=0. &\text{(A.9c)}
\end{aligned}$$

Combining Eq. (A.8) and Eq. (A.9) with the operation (A.8)+$\varepsilon\times$(A.9), we obtain

$$\begin{aligned}
&\partial_t u_x+\partial_x(p+u_x^2)+\partial_y(u_xu_y)+\partial_z(u_xu_z)\\
&=-\varepsilon\partial_y[(1-\lambda_\nu'/2)p_{xy}^{(1)}]-\varepsilon\partial_x[(1-\lambda_\nu/2)S_{xx}^{(1)}+2(1-\lambda_e/2)e^{(1)}/39]\\
&\quad-\varepsilon\partial_z[(1-\lambda_\nu'/2)S_{xz}^{(1)}], &\text{(A.10a)}\\
&\partial_t u_y+\partial_x(u_xu_y)+\partial_y(p+u_y^2)+\partial_z(u_yu_z)\\
&=-\varepsilon\partial_x[(1-\lambda_\nu'/2)p_{xy}^{(1)}]-\varepsilon\partial_y[(1-\lambda_\nu/2)(S_{\omega\omega}^{(1)}-S_{xx}^{(1)})/2+2(1-\lambda_e/2)e^{(1)}/39]\\
&\quad-\varepsilon\partial_z[(1-\lambda_\nu'/2)S_{yz}^{(1)}], &\text{(A.10b)}\\
&\partial_t u_z+\partial_x(u_xu_z)+\partial_y(u_yu_z)+\partial_z(p+u_z^2)\\
&=-\varepsilon\partial_x[(1-\lambda_\nu'/2)S_{xz}^{(1)}]-\varepsilon\partial_y[(1-\lambda_\nu'/2)S_{yz}^{(1)}]-\varepsilon\partial_z[2(1-\lambda_e/2)e^{(1)}/39\\
&\quad-(1-\lambda_\nu/2)(S_{\omega\omega}^{(1)}+S_{xx}^{(1)})/2]. &\text{(A.10c)}
\end{aligned}$$

According to Eq. (A.6), yields

$$\begin{aligned}
&e^{(1)}=-\frac{1}{\lambda_e}[\partial_{t_0}(13u^2/2+39p/2-12\rho_0)+(\partial_x u_x+\partial_y u_y+\partial_z u_z)], &\text{(A.11a)}\\
&S_{xx}^{(1)}=-\frac{1}{3\lambda_\nu}\left[\partial_{t_0}(3u_x^2-u^2)+\partial_x u_x-\frac{1}{2}\partial_y u_y-\frac{1}{2}\partial_z u_z\right], &\text{(A.11b)}
\end{aligned}$$

$$S_{\omega\omega}^{(1)} = -\frac{1}{\lambda_\nu}\Big[\partial_{t_0}(u_y^2 - u_z^2) + \frac{1}{2}\partial_y u_y - \frac{1}{2}\partial_z u_z\Big], \tag{A.11c}$$

$$S_{xy}^{(1)} = -\frac{1}{\lambda'_\nu}\Big[\partial_{t_0}(u_x u_y) + \frac{1}{2}\partial_x u_y + \frac{1}{2}\partial_y u_x\Big], \tag{A.11d}$$

$$S_{yz}^{(1)} = -\frac{1}{\lambda'_\nu}\Big[\partial_{t_0}(u_y u_z) + \frac{1}{2}\partial_y u_z + \frac{1}{2}\partial_z u_y\Big], \tag{A.11e}$$

$$S_{xz}^{(1)} = -\frac{1}{\lambda'_\nu}\Big[\partial_{t_0}(u_x u_z) + \frac{1}{2}\partial_x u_z + \frac{1}{2}\partial_z u_x\Big]. \tag{A.11f}$$

Using Eq. (A.8), we can estimate the terms $\partial_{t_0} u_x^2$, $\partial_{t_0} u_y^2$ and $\partial_{t_0} u_z^2$ as

$$\partial_{t_0} u_x^2 = -2u_x(\partial_x(p+u_x^2) + \partial_y(u_x u_y) + \partial_z(u_x u_z)), \tag{A.12a}$$

$$\partial_{t_0} u_y^2 = -2u_y(\partial_x(u_x u_y) + \partial_y(p+u_y^2) + \partial_z(u_y u_z)), \tag{A.12b}$$

$$\partial_{t_0} u_z^2 = -2u_z(\partial_x(u_x u_z) + \partial_y(u_y u_z) + \partial_z(p+u_z^2)). \tag{A.12c}$$

From Eq. (A.1), we can see that $\partial_{t_0} u_x^2$, $\partial_{t_0} u_y^2$ and $\partial_{t_0} u_z^2$ are all in the order of $\mathcal{O}(M^3)$. Omitting the $O(M^3)$ terms, Eq. (A.11) becomes

$$e^{(1)} = -\frac{39}{2\lambda_e}\partial_{t_0} p, \tag{A.13a}$$

$$S_{xx}^{(1)} = -\frac{1}{6\lambda_\nu}(2\partial_x u_x - \partial_y u_y - \partial_z u_z) = -\frac{1}{2\lambda_\nu}\partial_x u_x, \tag{A.13b}$$

$$S_{\omega\omega}^{(1)} = -\frac{1}{2\lambda_\nu}(\partial_y u_y - \partial_z u_z), \tag{A.13c}$$

$$S_{xy}^{(1)} = -\frac{1}{2\lambda'_\nu}(\partial_x u_y + \partial_y u_x), \tag{A.13d}$$

$$S_{yz}^{(1)} = -\frac{1}{2\lambda'_\nu}(\partial_y u_z + \partial_z u_y), \tag{A.13e}$$

$$S_{zx}^{(1)} = -\frac{1}{2\lambda'_\nu}(\partial_x u_z + \partial_z u_x), \tag{A.13f}$$

where the continuity equation has been substituted into Eq. (A.13a) and Eq. (A.13b), and the term $\partial_{t_0} u^2$ in Eq. (A.13a) has been omitted.

Substituting Eq. (A.13) into Eq. (A.10) and supposing

$$\varepsilon = \delta_t, \quad \nu = \frac{1}{4}(1/\lambda_\nu - 1/2)\delta_t = \frac{1}{2}(1/\lambda'_\nu - 1/2)\delta_t,$$

then we have

$$\partial_t u_x + \partial_x(u_x^2) + \partial_y(u_x u_y) + \partial_z(u_x u_z) = -\partial_x p + \nu(\partial_x^2 u_x + \partial_y^2 u_x + \partial_z^2 u_x), \tag{A.14a}$$

$$\partial_t u_y + \partial_x(u_x u_y) + \partial_y(u_y^2) + \partial_z(u_y u_z) = -\partial_y p + \nu(\partial_x^2 u_y + \partial_y^2 u_y + \partial_z^2 u_y), \tag{A.14b}$$

$$\partial_t u_z + \partial_x(u_x u_z) + \partial_y(u_y u_z) + \partial_z(u_z^2) = -\partial_z p + \nu(\partial_x^2 u_z + \partial_y^2 u_z + \partial_z^2 u_z), \tag{A.14c}$$

which are the momentum equations of the incompressible N-S equations. It should be noted that the continuity equation has been used and the term $\varepsilon\partial_\alpha\partial_t p$ $(\alpha = x,y,z)$ of order $\mathcal{O}({\delta_t}^2 + \delta_t M^2)$ is omitted in deriving the above equations.

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